

Hospital Readmission Analysis

Executive Report

Prepared from hospital_readmission_dataset.xlsx • 8,010 patient records

1. Executive Summary

This report summarizes the readmission-risk profile of 8,010 hospital admissions across five regions, eleven primary diagnoses, and four insurance categories. The dataset shows an average length of stay of 7.8 days, an average patient age of roughly 57.5 years, and an average predicted readmission risk score of 0.78 out of 1.0 — with 6,192 of 8,010 admissions (77%) carrying the model's positive readmission label. Diabetes is the most common diagnosis by volume, but Heart Failure, COPD, Sepsis, and Stroke are the diagnoses driving the highest average readmission risk and longest average stays, which is where clinical and care-coordination resources will have the most impact.

2. Key Findings

2.1 Admissions by diagnosis

Diabetes accounts for the largest share of admissions, followed by Hypertension and COPD. Together, the top three diagnoses make up nearly half of all admissions.

Diagnosis	Admissions	Share of total
Diabetes	1,472	18.4%
Hypertension	1,303	16.3%
COPD	953	11.9%
Stroke	885	11.0%
Kidney Disease	868	10.8%
Appendicitis	535	6.7%
Sepsis	497	6.2%
Fracture	434	5.4%
Pneumonia	426	5.3%
Heart Failure	404	5.0%
Influenza	233	2.9%

2.2 Admissions by region

The South region records the highest admission volume, roughly 30% more than the lowest-volume region (Central), suggesting either higher regional demand or a larger catchment population served there.

Region	Admissions
South	1,994
West	1,631
East	1,617
North	1,525
Central	1,235

2.3 Length of stay by diagnosis

Sepsis drives the longest average stay by a clear margin, consistent with its clinical severity, followed by Stroke and Heart Failure. Hypertension and Appendicitis have the shortest average stays.

Diagnosis	Avg. length of stay (days)
Sepsis	10.8
Stroke	9.7
Heart Failure	9.7
COPD	8.8
Pneumonia	8.7
Fracture	7.8
Kidney Disease	7.7
Influenza	6.8
Diabetes	6.8
Appendicitis	5.9
Hypertension	5.8

2.4 Patients by insurance type

Privately insured patients make up the largest payer group, with Medicare close behind — together accounting for about 70% of all admissions. Uninsured patients are the smallest group but still represent over 800 admissions worth of uncompensated-care exposure.

Insurance type	Patients	Share of total
Private	3,053	38.2%
Medicare	2,543	31.8%
Medicaid	1,597	20.0%
Uninsured	809	10.1%

2.5 Average readmission risk by diagnosis

Heart Failure carries the highest average predicted readmission risk, tightly clustered with COPD, Sepsis, Stroke, and kidney disease — all chronic or acute conditions known for high recidivism. Notably, these are not the diagnoses with the highest admission volume: Diabetes and Hypertension are common but comparatively lower-risk on this measure.

Diagnosis	Avg. readmission risk score
Heart Failure	0.88
COPD	0.87
Sepsis	0.85
Stroke	0.85
Kidney Disease	0.85
Influenza	0.74
Appendicitis	0.72
Pneumonia	0.72
Fracture	0.72
Hypertension	0.70
Diabetes	0.70

3. Data Quality Notes

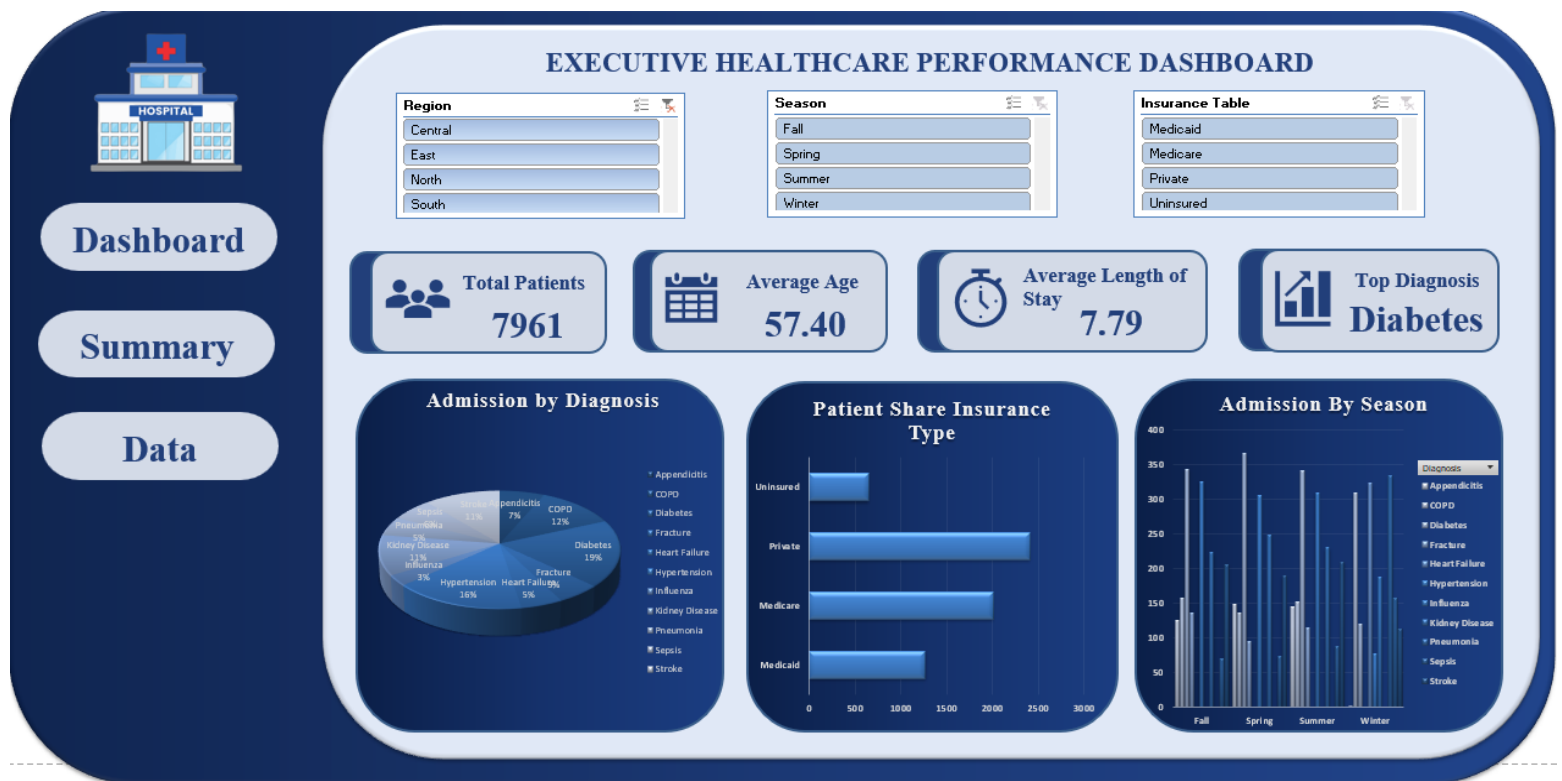
A handful of data quality issues were identified during analysis and should be corrected before the figures above are used for clinical decision-making (the accompanying Power Query steps in the project workbook resolve all of these):

- 18 records store age as text with stray characters (e.g. '70, \$42, 57 kg) rather than a clean number.
- A small number of age values are implausible outliers (including a negative value and a value of 250), indicating data-entry errors.
- 21 patient_id values appear more than once in the raw extract.
- 8 rows are missing patient_id, admission_date, and season.
- Diagnosis and insurance codes in the two lookup tables carry a leading space, which must be trimmed before they will match the codes in the main data.

4. Executive Recommendations

- Evaluate capacity and staffing in the South region - they have the largest share of admissions in isolation.
- Analyze bed utilization and discharge workflow in Heart Failure, Sepsis, and Stroke as they account for lengthiest LOS.
- Develop and expand follow-up care coordination and post discharge services for all heart failure, COPD, Sepsis, stroke, and Kidney Disease patients - These are the five largest categories by number of readmissions even if some have fewer cases.

5. Executive Dashboard



6. LinkedIn Profile

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